Project Design Phase-II
Technology Stack (Architecture & Stack)

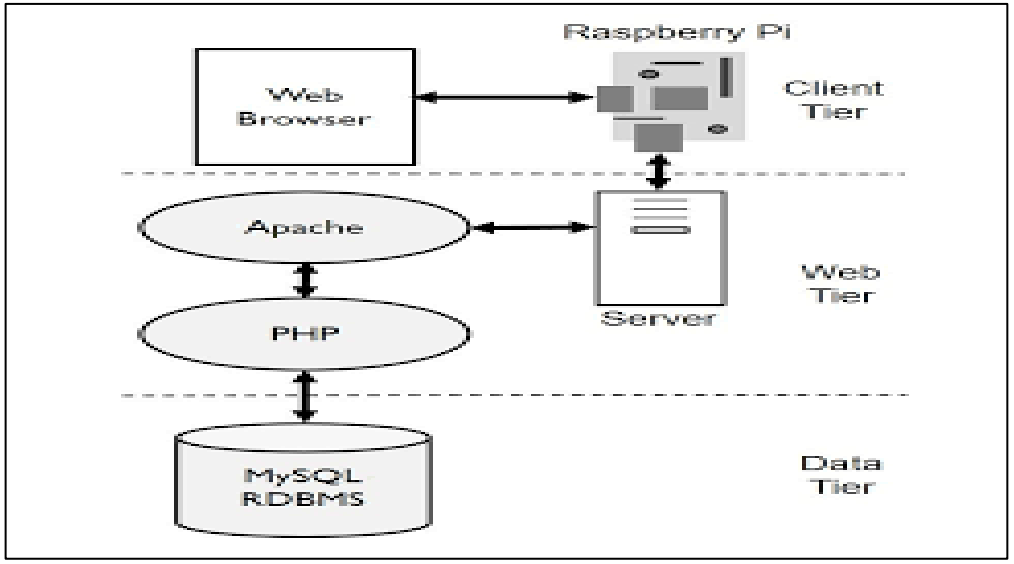
Date	27 FEB 2026
Team ID	LTVIP2026TMIDS41352
Project Name	ShopEZ:One-Stop Shop for Online Purchases
Maximum Marks	4 Marks

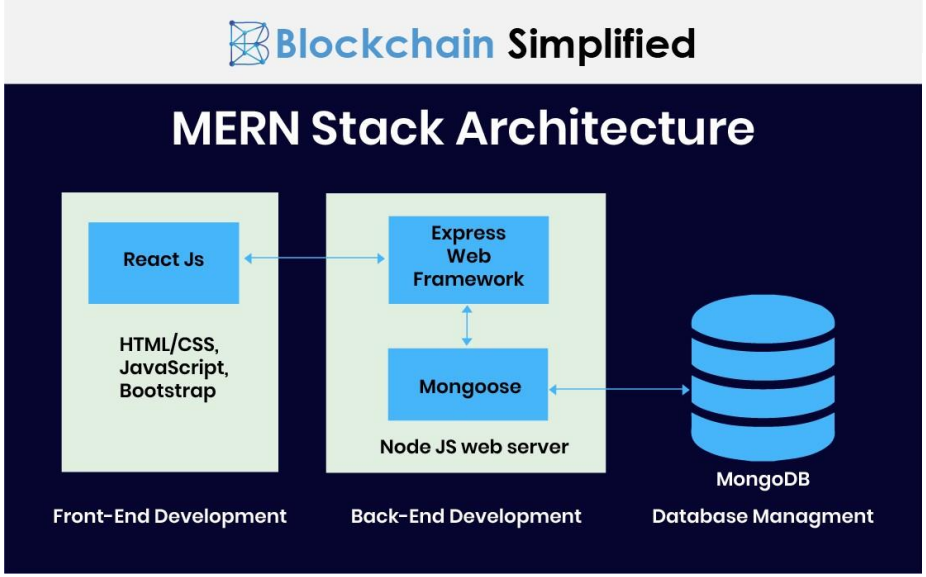
Technical Architecture:

ShopEZ follows a 3-Tier MERN Architecture consisting of:

1. **Presentation Layer (Frontend)**
2. **Application Layer (Backend APIs)**
3. **Data Layer (Database)**

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/





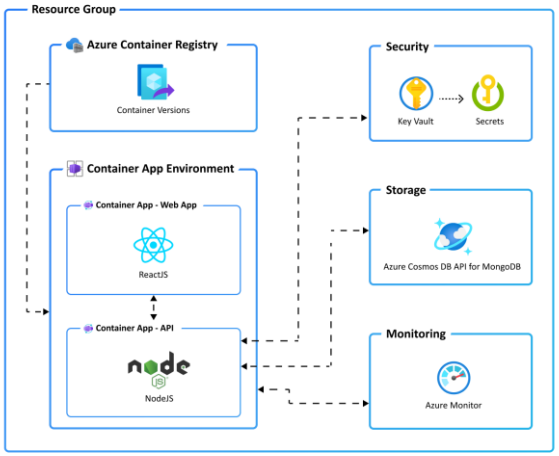


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based interface for customers, sellers, and admin	HTML, CSS, Bootstrap, React JS
2	Application Logic-1	User authentication & authorization	Node.js, Express.js, JWT
3	Application Logic-2	Product & Cart management	Node.js, Express.js
4	Application Logic-3	Order processing & seller dashboard	Node.js, Express.js
5	Database	Stores users, products, orders, cart data	MongoDB (NoSQL)
6	Cloud Database	Managed database service	MongoDB Atlas
7	File Storage	Product images storage	Cloudinary / Local File System
8	External API-1	Payment Integration	Razorpay / Stripe API
9	External API-2	Email Notification Service	Nodemailer / SendGrid
10	Machine Learning Model	(Future Scope) Product Recommendation	Basic Recommendation Logic / ML Model
11	Infrastructure	Application Deployment	Render / Vercel / AWS / Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Frameworks used for development	React JS, Node.js, Express.js, MongoDB
2	Security Implementations	Secure authentication and data protection	JWT, Bcrypt Password Hashing, HTTPS, Role-Based Access

S.No	Characteristics	Description	Technology Used
3	Scalable Architecture	3-Tier Architecture supporting horizontal scaling	MERN Stack, Cloud Hosting
4	Availability	24/7 availability via cloud deployment	MongoDB Atlas, Cloud Hosting
5	Performance	Optimized API calls and fast page rendering	React Virtual DOM, Indexed Queries, Caching
6	Maintainability	Modular folder structure for frontend & backend	MVC Architecture Pattern
7	Reliability	Proper error handling & validation	Express Middleware Validation

References:

- **C4 Model for Software Architecture.**
  Available at: https://c4model.com/
  Accessed for understanding system context, container, and component-level architectural diagrams.

- **IBM Developer – Online Order Processing System During Pandemic.**
  Available at: https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
  Referenced for architectural pattern inspiration and scalable backend design.

- **IBM Cloud Architecture Center.**
  Available at: https://www.ibm.com/cloud/architecture
  Used for understanding cloud-native application deployment models.

- **AWS Architecture Center.**
  Available at: https://aws.amazon.com/architecture
  Referenced for best practices in scalable and distributed system design.

Medium – How to Draw Useful Technical Architecture Diagrams.
Available at: https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d
Used as guidance for structuring architecture diagrams.

MongoDB Official Documentation.
Available at: https://www.mongodb.com/docs/
Referenced for database schema design and configuration.

React.js Official Documentation.
Available at: https://react.dev/
Used for frontend component architecture and state management understanding.

Node.js Official Documentation.
Available at: https://nodejs.org/en/docs/
Referenced for backend server setup and API handling.

Express.js Documentation.
Available at: https://expressjs.com/
Used for REST API development and middleware implementation.

JSON Web Token (JWT) Documentation.
Available at: https://jwt.io/
Referenced for authentication implementation.